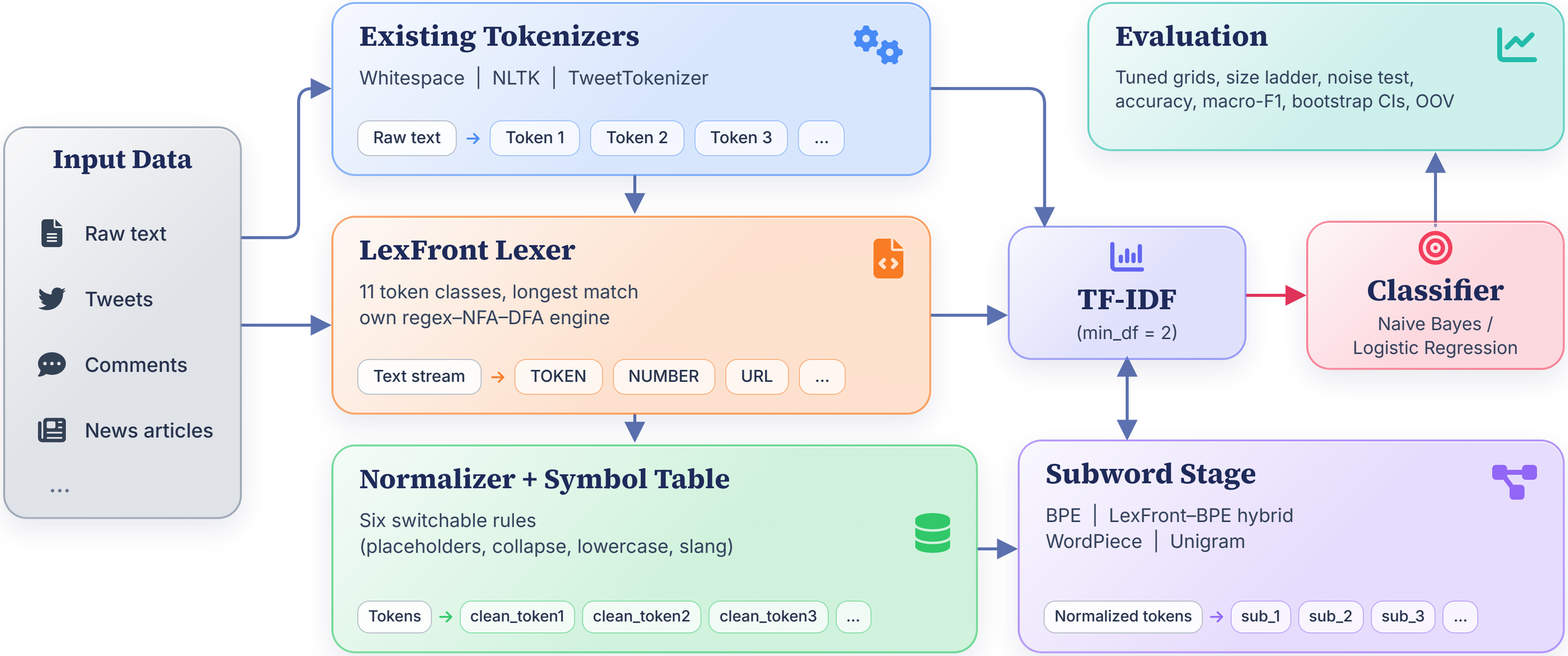

Input Data
Raw text
Tweets
Comments
News articles
...
Existing Tokenizers
Whitespace | NLTK | TweetTokenizer
Raw text
Token 1
Token 2
Token 3
...
LexFront Lexer
11 token classes, longest match
own regex–NFA–DFA engine
Text stream
TOKEN
NUMBER
URL
...
Normalizer + Symbol Table
Six switchable rules
(placeholders, collapse, lowercase, slang)
Tokens
clean_token1
clean_token2
clean_token3
...
TF-IDF
(min_df = 2)
Subword Stage
BPE | LexFront–BPE hybrid
WordPiece | Unigram
Normalized tokens
sub_1
sub_2
sub_3
...
Classifier
Naive Bayes /
Logistic Regression
Evaluation
Tuned grids, size ladder, noise test,
accuracy, macro-F1, bootstrap CIs, OOV